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Q.1 Implementation of stack using Array.

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
#include<string.h>
```

```
#define Max 50
```

```
int stack[Max];
```

```
int top=-1;
```

```
void pop();
```

```
void push();
```

```
void display()
```

```
{
```

```
int n;
```

```
if(stack[top]==Max-1)
```

```
printf("stack overflow");
```

```
else{ printf("enter data");
```

```
scanf("%d",&n);
```

```
top++;
stack[top]=n;
}
}
void pop()
{
if(stack[top]==- 1)
printf("stack under flow");
else{ printf("deleted element is %d",stack[top-- ]);
}
}
void display()
{
if(top== - 1)
{
printf("stack underflow");
}
Else
{
for(int i=top;i>=0;i--)
```

```
{  
    printf("%d \n",stack[i]);  
}  
}  
}  
  
int main()  
{  
    int choice;  
    while(1)  
    {  
        printf("select your choice \n 1.push \n 2.pop  
 \n3.display \n4. exit");  
        scanf("%d",&choice);  
        switch(choice)  
        {  
            case 1:push();  
            break;  
            case 2:pop();  
            break;  
            case 3:display();
```

```
break;  
    case 4:exit(0);  
    }  
}  
return 0;  
}
```

Output:

select your choice

1.push

2.pop

3.display

4.exit

1

enter data10

select your choice

1.push

2.pop

3.display

4.exit

1

enter data20

select your choice

1.push

2.pop

3.display

4.exit

3

20 10

select your choice

1.push

2.pop

3.display

4.exit

2

deleted element is 20

select your choice

1.push

2.pop

3.display

4.exit

3

10

select your choice

1.push

2.pop

3.display

4.exit

Q.2 Implementation of Stack using Linked List.

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
struct node
```

```
{
```

```
int data;
```

```
struct node *next;
```

```
};
```

```
struct node *start=NULL;
```

```
void push()
```

```
{
```

```
struct node *temp,*ptr=start;
```

```

int num;

printf("\n enter the data");
scanf("%d",&num);
temp=(struct node *)malloc(sizeof(struct node));
temp->data=num;
if(start==NULL)
{ start=temp;
temp->next=NULL;
}
Else
{
if(start->next==NULL)
{
start->next=temp;
temp->next=NULL;
}
Else
{
while(ptr->next!=NULL)
{

```

```
ptr=ptr->next;
}
ptr->next=temp;
temp->next=NULL;
}
}
printf("inserted no. is =%d",temp->data);
}
void pop()
{
    struct node *temp=start,*ptr;
    if(start==NULL)
        printf("\n the stack underflow");
    else {
        if(start->next==NULL)
        {
            printf("\n deleted element is %d",temp->data);
            start=NULL;
            free(temp);
        }
    }
}
```



```

while(temp->next!=NULL)
{
    ptr=temp;
    temp=temp->next;
}
printf("\n deleted element is %d",temp->data);
ptr->next=NULL;
free(temp);
}
}

void display()
{
    struct node *temp=start;
    if(start==NULL)
        printf("\n stack underflow");
    else{
        //struct node
        *temp=start;
        printf("\n the elements in stack are :");
        while(temp!=NULL)

```

```
{  
    printf("\n %d",temp->data);  
    temp=temp->next;  
}  
  
}  
  
}  
  
int main()  
{  
    printf("enter your choice");  
    int choice,loop=1;  
    while(1)  
    {  
        printf("\n 1.push\n 2.pop\n 3.display\n ");  
        scanf("%d",&choice);  
        switch(choice)  
        {  
            case 1:push();  
            break;  
            case 2:pop();  
            break;
```

```
case 3:display();  
break;  
case 4:exit(0);  
break;  
}  
}  
return 0;  
}
```

Output:

enter your choice

1.push

2.pop

3.display

1

enter the data10

inserted no. is =10

1.push

2.pop

3.display

1

enter the data 20

inserted no. is = 20

1.push

2.pop

3.display

3

the elements in stack are : 10 20

1.push

2.pop

3.display

2

deleted element is 20

1.push

2.pop

3.display

3

the elements in stack are : 10

1.push

2.pop

3.display